



Contents lists available at ScienceDirect

Journal of Experimental Child Psychology

journal homepage: www.elsevier.com/locate/jecp



Impacts of number lines and circle visual displays on caregivers' fraction understanding



Daniel A. Scheibe^{a,*}, Lauren Wyatt^a, Charles J. Fitzsimmons^b,
Marta K. Mielicki^{c,d}, Lauren K. Schiller^e, Clarissa A. Thompson^a

^a Department of Psychological Sciences, Kent State University, Kent, OH 44240, USA

^b Department of Psychology, University of North Florida, Jacksonville, FL 32224, USA

^c Psychological Sciences, Rutgers University, New Brunswick, NJ 08901, USA

^d SRI International, Menlo Park, CA 94025, USA

^e Teachers College, Columbia University, New York, NY 10027, USA

ARTICLE INFO

Article history:

Available online xxxx

Keywords:

Fractions

Arithmetic

Equivalence

Math anxiety

Picture books

Back matter

ABSTRACT

Playful fraction picture books, together with math instructional content called “back matter,” may promote fraction learning, which is crucial because fractions are difficult and often disliked content. However, open questions remain regarding how different types of back matter may affect caregivers' ability to use fraction picture books as a teaching tool. The current study offers a novel investigation into how back matter affects caregivers' ($N = 160$) fraction understanding (i.e., equivalence and arithmetic) and subjective beliefs about math using a pretest/posttest design. We contrasted existing back matter text with research-informed back matter text crossed with either circle area or number line visual displays. Caregivers' performance improved from pretest to posttest in the Researcher-Generated + Circles condition (fraction equivalence) and in the Existing + Circles, Researcher-Generated + Circles, and Researcher-Generated + Number Lines conditions (fraction arithmetic). In addition, caregivers were aware of their learning; they predicted improvements in their fraction arithmetic performance over time. These findings suggest that brief interventions, such as back matter in children's picture books, may improve adults' fraction understanding.

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* Corresponding author.

E-mail address: dscheib2@kent.edu (D.A. Scheibe).

Introduction

Fractions are an essential component of elementary math understanding. The mastery of fraction concepts can predict success in more advanced math areas, especially those that require a background in algebra (Bailey et al., 2012). Yet, fractions are not always prioritized in the informal home learning environment (cf. Eason & Ramani, 2020). This may be because children's and adults' previous experiences in interacting with fractions in a formal math context (i.e., the classroom) invoked fear or boredom (Luttenberger et al., 2018; Sidney et al., 2021), negatively affecting the frequency with which fractions are discussed in the home. In instances when caregivers engage children in fraction learning in the home, it is not clear whether those methods are as effective as they could be. One tool caregivers can use to aid in working with fractions in the home is children's books (Purpura et al., 2021). Indeed, caregivers have access to many different children's books that incorporate fraction knowledge (i.e., fraction picture books). Recent research found that parents' use of numerical language in the context of reading books with their children may have a causal impact on improving their children's number knowledge (Gibson et al., 2020). Children's books can sometimes include instructions and visual displays—referred to as *back matter*. Back matter is included to help caregivers support children as they learn about fractions and can include visual displays such as number lines and circle area models. The goal of this research is to equip caregivers with more knowledge about fractions as part of activities they are already doing with their children to potentially improve the quality of learning activities in the home. The current study addressed different types of back matter informed by the integrated theory of numerical development (e.g., the same number on a number line can be represented in an infinite number of ways; Schiller & Siegler, 2023; Siegler et al., 2011). The aim was to focus on one aspect of children's picture books by incorporating empirical findings from math cognition as a way to improve caregivers' fraction knowledge.

Despite the fact that many fraction picture books including back matter exist, the efficacy of existing back matter at improving fraction understanding is understudied. Indeed, it is unclear to what extent the back matter was inspired by findings from math cognition and math education research. Open questions remain regarding how different forms of back matter in currently available fraction books may affect their usefulness as a learning aid. The current study investigated whether researcher-generated, theory-informed back matter text (hereafter RGT for researcher-generated text) may be more effective at helping caregivers understand fractions than existing back matter text (hereafter ET for existing text) in children's fraction books.

The primary research questions of the current study involved evaluating the independent and combined effects of back matter text (RGT or ET) and visual displays (circle area models or number lines; see Fig. 1) at improving adult caregivers' fraction equivalence (e.g., $1/2 = 15/30$) and fraction arithmetic (e.g., $2/7 + 3/5$) performance. We also evaluated whether the pairings of different types of back matter text with different visual displays had an influence on caregivers' ratings of math anxiety, confidence in their fraction arithmetic problem-solving abilities, and preferences about children's fraction picture books. These are important research questions because children and adults struggle to comprehend fractions (Thompson et al., 2023) and report disliking these numbers (Mielicki et al., 2022; Sidney et al., 2021). Furthermore, there is some evidence of transmission of math anxiety from parents to their children (e.g., Maloney et al., 2015). Therefore, efforts should be undertaken to promote more positive attitudes about this consequential type of math as parents and children interact with books in the home environment. It is possible that carefully curated back matter could increase parents' confidence in their own understanding of this difficult math content and help them to feel empowered when they encounter fraction topics in informal learning contexts such as when completing homework and reading books together.

The importance of learning about fractions at home

Children typically learn about fractions in elementary school, yet they also learn about fractions outside of the classroom. The extent to which caregivers engage children in informal home math activ-

Existing Back Matter + Circle Visual

$$\frac{1}{5} + \frac{2}{5} = \frac{3}{5} \quad (\text{The LCD is 5.})$$


Displays

Researcher-Generated Back Matter +

Circle Visual Displays



$$\frac{1}{5} + \frac{2}{5} = \frac{3}{5}$$

Existing Back Matter + Number Line Visual

$$\frac{1}{5} + \frac{2}{5} = \frac{3}{5} \quad (\text{The LCD is 5.})$$


Displays

Researcher-Generated Back Matter + Number

Line Visual Displays

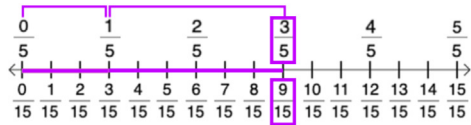
$$\frac{1}{5} + \frac{2}{5} = \frac{3}{5}$$


Fig. 1. Examples of visuals for each experimental condition. LCD, lowest common denominator.

ities may benefit children's early numerical understanding (Berkowitz et al., 2015; Kleemans et al., 2012; LeFevre et al., 2009; Purpura et al., 2021; Skwarchuk et al., 2014). Caregivers play a crucial role in helping children to expand their knowledge of fractions given that both the quantity and quality of math interactions between caregivers and children influence children's learning (Ramani et al., 2015). The role of caregivers in children's learning was heightened during the recent COVID-19 pandemic when school-aged children were spending less time in traditional classroom settings and more time at home with their caregivers (Thompson, Schiller, et al., 2021). Children not only learn in formal classroom settings but also engage in informal learning experiences with their caregivers, for instance, when caregivers and their children bake together and use one fourth of a cup of sugar or when caregivers tells their children that the time is "a quarter past one." Other times, caregivers deliberately discuss fractions with their children in the home environment. For instance, some caregivers might include fractions in their dialogue with their children by reading picture books to them that highlight fraction concepts. Recent research (e.g., Purpura et al., 2021) demonstrated that when caregivers read picture books with mathematical content to their children, there may be positive effects on children's math learning (e.g., numeracy).

One way that fraction picture books foster children's learning about fractions is by including instructions (i.e., back matter) that unpack fraction concepts central to the book using text, visual displays, or both. In this way, back matter is akin to a worked example (McGinn et al., 2015) in which a math problem is shown being solved step by step in an effort to alleviate misconceptions. A recent meta-analysis indicated that worked examples have a medium positive effect on math performance (Barbieri et al., 2023). Given the similarity between worked examples and back matter, back matter

may be similarly beneficial for improving math performance. Back matter sets the stage for how fractions should be understood in the context of the book. It may be especially useful for helping children with learning disabilities given research suggesting that these children benefit from interventions composed of explicit math instruction (Gersten et al., 2009). Back matter may also aid caregivers by providing a refresher on fraction concepts that they have not encountered recently, thereby providing caregivers with tools to teach their children more effectively about fractions.

Back matter is a tool used in many children's picture books, not only math books, as a way to provide additional or explanatory information about the topics in the books. (See <https://www.wordsandpics.org/2019/10/picture-book-focus-creating-compelling.html> for tips from an editor on creating back matter for nonfiction picture books.) Within fraction picture books, back matter can serve as a quick and engaging lesson on, or a reminder of, key fraction concepts used within the books. This can help children and caregivers to understand how fractions are used within the book and familiarize them with key terms that might be used throughout the story.

Caregivers' feelings about math matter

Although it is becoming increasingly evident that the home learning environment is a key component of children's learning (Gibson et al., 2020; Hornburg et al., 2021; Purpura et al., 2021), many adults endorse negative attitudes toward math (Mielicki et al., 2022; Sidney et al., 2021) or experience math anxiety (Ashcraft, 2002; Hart & Ganley, 2019). Caregivers' attitudes toward and anxiety about math are important because when negativity toward math is modeled in front of children, it can negatively affect the home math environment (Berkowitz et al., 2021; Missall et al., 2015; Schaeffer et al., 2018; Skwarchuk et al., 2014) because children's math self-efficacy is at least partly determined by external feedback from caregivers (Becker et al., 2022; Jameson, 2014; Lee et al., 2022; Maloney et al., 2015). In addition, caregivers may intergenerationally transmit their own math anxiety to their children (Casad et al., 2015; Herts et al., 2019; Maloney et al., 2015). Thus, it is important that caregivers—especially those who report negative math attitudes or math anxiety and those who have negative perceptions of their own math ability—have effective tools to assist them in teaching their children in the home learning environment. Indeed, recent research suggests that young children might not be encouraged to think as deeply about math concepts as literacy concepts (Uscianowski et al., 2020), perhaps because most caregivers have more positive feelings toward reading than toward math (Maloney et al., 2015).

Caregivers' confidence (i.e., metacognitive judgments about their own understanding of the content caregivers are teaching to their children) when using specific tools (e.g., fraction picture books) to teach children about math might also predict the extent to which such tools are used in the home environment. For example, if caregivers report being more confident in their relevant math abilities as they are in reading fraction picture books, they may be more likely to spend more time using fraction picture books as a tool to aid their children's learning. This behavioral impact would be important because regularly reading fraction picture books to children can have strong positive effects on children's mathematical language and numeracy skills (Purpura et al., 2021). However, open questions remain regarding what type of content fraction books should contain—in terms of both text and visual displays.

To begin addressing the open questions regarding fraction books, our team conducted a search of fraction books currently easily accessible by caregivers. We found that back matter is not always included in picture books; roughly 43% of the fraction books we analyzed did *not* include back matter (see Table S1 in the online [supplementary material](#) for an overview of 21 surveyed fraction books and an analysis of existing back matter in these books).¹ Some of the authors of fraction books are schoolteachers who wrote the books to better teach their students (i.e., the author of *Piece = Part = Portion*). Their way of writing the content and back matter may be influenced by what they have seen work

¹ We do not propose that the 21 books presented in Table S1 represent an exhaustive list of all fraction books currently in print; however, these books represent a sample of those that are readily accessible in print, books that caregivers are likely to come across in the library or purchase in a bookstore. Indeed, we consulted with a librarian and conducted an independent investigation to try and find a sample of fraction books available in the United States.

in their own classrooms. We cannot conclusively say how authors prepared the back matter given that within the books we analyzed the authors did not detail their process for creating their back matter or the educational sources they consulted.

Do visual displays influence fraction learning and math anxiety?

Although circle visual displays (see Fig. 1 for examples) are prevalent in picture books and in math instruction, they are potentially problematic as a means for representing fraction magnitudes. Representing fractions as circles, as opposed to using number lines, can lead to lower performance on fraction estimation and comparison for children (Hamdan & Gunderson, 2017) as well as fraction arithmetic for children (Gunderson et al., 2019; Sidney et al., 2019) and adults (Sidney et al., 2022). One hypothesis is that number lines are often more effective than circle displays in that they align with the proposed mental number line (Dehaene, 2011) because smaller magnitudes are represented on the left side of space and larger magnitudes are represented on the right side of space (Opfer et al., 2010). Number lines may also promote a clearer comparison of fraction magnitudes and an understanding of fractions as units of measurement (i.e., distance) in addition to “parts of a whole” (Moss & Case, 1999). Indeed, interventions featuring number lines (Fuchs et al., 2016) improved children’s understanding of fraction magnitudes, especially for children at risk of poor math performance (see also Bustamante et al., 2022). Other work also found benefits of number lines for training adults’ fraction magnitude understanding in COVID-19 health risk contexts (Fitzsimmons et al., 2023; Mielicki et al., 2023; Thompson, Taber, et al., 2021).

Although presenting fractions as number lines may benefit fraction learning and performance on fraction tasks, caregivers may experience more subjective feelings of ease when working with circle visual displays. Furthermore, they might find circle visual displays helpful for reasoning about fraction magnitudes because of their familiarity with fractions represented as circle displays during their own formal schooling (Sidney et al., 2019, 2022). It was not until 2010 that American schools began adopting the Common Core State Standards for Mathematics. One of the standards for third-grade math is that fractions should be taught as a number on the number line (Grade 3 >> Numbers & Operations—Fractions, 2010). Because these standards were not in place when adults were first learning about fractions during elementary school, they primarily received fraction instruction involving area models, such as circles and rectangles, when they learned about fractions (Sidney et al., 2019, 2022). This recent development in teaching fractions highlights intergenerational differences in the ways fractions were previously taught (i.e., circle visual displays) and improved strategies for scaffolding fraction knowledge based on more recent research in math cognition (e.g., primarily using number lines; Siegler et al., 2011). The intergenerational differences may help to explain why there are so few children’s books that exist with fractions displayed using a number line. Thus, we tested the role of caregivers’ subjective feelings about both circle visual displays and number line visual displays in the current study.

Does back matter phrasing influence fraction understanding?

In addition to differences in back matter visual displays (e.g., circles or number lines), it is important to examine word choice in back matter content for at least two reasons. First, word choice could affect quantity representation. For instance, some languages (i.e., Korean) include fraction naming conventions that are associated with children’s deeper fraction understanding. When English-speaking children heard English translations of the Korean wording for fractions (i.e., “one of four parts”), these children were significantly better at matching the fraction words to circle and rectangle visual displays than English-speaking children who heard traditional English fraction names (i.e., “one fourth”) (Mix & Paik, 2008; Paik & Mix, 2003). Korean fraction labels may help to emphasize the part-whole structure of fractions because they explicitly mention fractional parts (e.g., “one of four parts”), whereas English fraction labels tend to imply fractional parts more implicitly (e.g., “one fourth”). Labeling fractions to make it easier to identify the numerator and denominator may facilitate deeper conceptual understanding of fractions (Rittle-Johnson, 2017).

In some existing children's books, the back matter phrasing does not adhere to evidenced-based best practices and could potentially be preventing children and adults from optimally learning about and understanding fractions. For instance, the back matter wording from one published children's fraction book, *Fractions in Disguise* (see Appendix A), lacks the phrasing about parts and wholes that could aid children in effectively comprehending how numerators relate to denominators and result in the fractions' magnitudes. In fact, the back matter for *Fractions in Disguise* included complex mathematical vocabulary, such as "greatest common factor" and "lowest common denominator," which could be confusing or anxiety-provoking for caregivers who are far removed from learning about math in formal classroom environments. By empirically comparing this existing back matter phrasing with phrasing more aligned with empirical evidence, it may be possible to evaluate how future back matter should be incorporated into children's fraction picture books.

The second reason to consider word choice in back matter is that it could have an effect on math anxiety (Ashcraft, 2002; Ramirez et al., 2013). For those who experience math anxiety, interacting with math can lead to feelings of panic and tension, which may then negatively influence mathematical performance (Ashcraft, 2002, 2019; Ashcraft & Ridley, 2005; Luttenberger et al., 2018; Ramirez et al., 2018). Feelings of anxiety toward math can be exacerbated when individuals encounter mathematical concepts they are not familiar or comfortable with or that require previous domain knowledge they are lacking (Finlayson, 2014; Sidney & Thompson, 2019). It may be helpful to define novel words (e.g., lowest common denominator, numerator, denominator; see Appendix A) and avoid unnecessary mathematical terminology in back matter (e.g., reducer; see Appendix A) to help caregivers feel more confident and capable when interacting with the book.

The current study and hypotheses

In the current study, we compared *existing back matter text* (ET) from the book *Fractions in Disguise* with our *researcher-generated back matter text* (RGT), featuring explicit part-whole language and an explanation of how numerators and denominators relate to one another. We investigated whether RGT versus ET improved caregivers' accuracy in answering fraction equivalence (i.e., whether two fractions are the same magnitude or size) and fraction arithmetic problems (i.e., addition and subtraction). In addition to manipulating the type of back matter, we also manipulated whether the back matter featured number lines or circle area visual displays (Hamdan & Gunderson, 2017; Sidney et al., 2019, 2022). Thus, the current study employed a 2 (Back Matter Text: RGT vs. ET) $\times$ 2 (Visual Display Type: circles vs. number lines) fully crossed pretest-posttest design. This design allowed us to better understand the features of back matter that are important for improving adults' understanding of fraction content.

The reason we chose *Fractions in Disguise* as a representative example of existing back matter for our current study is that the book's back matter includes two crucial fraction concepts that we wanted to evaluate: fraction equivalence and fraction arithmetic. In addition, *Fractions in Disguise* includes elements that are common across many of the books' back matter, particularly its presentation of circle models to illustrate fractions.

We made two preregistered hypotheses (https://osf.io/ny5mf?view_only=c944251dea254064ac-fe481d54335f54 https://osf.io/ny5mf?view_only=3cb31eb0ebfe4d16ab9da69f5f54ee5e).²

Hypothesis 1

The first hypothesis (H1) was that researcher-generated back matter text (RGT) would lead to improved math accuracy relative to existing back matter (ET). More specifically, we expected participants who saw the RGT to show larger gains in both fraction equivalence and fraction arithmetic problems relative to participants who saw ET with either visual display.

We did not preregister whether we anticipated a main effect of number lines because we did not have a directional hypothesis for this effect. Even though number lines are beneficial for facilitating

² We also preregistered that we would explore the effect of back matter on math anxiety, but due to a coding error, math anxiety was not included at posttest and we could not evaluate this question.

rational number understanding for children (Yu et al., 2022) and adults (Fitzsimmons et al., 2023; Mielicki et al., 2022; Thompson, Taber, et al., 2021), the ET included several complex math vocabulary terms that may have captured participants' attention, thereby rendering the visual displays less helpful. In addition, circle visual displays may be more familiar to adult participants than number line visual displays (e.g., Sidney et al., 2022), which could reduce the probability of a main effect of visual display.

Hypothesis 2

The second hypothesis (H2) was there would be a three-way interaction among test time, type of back matter text, and type of visual display. We expected participants who saw the RGT paired with the number line visual display to show the largest improvements in fraction equivalence and fraction arithmetic accuracy from pretest to posttest after engaging with the back matter. See Fig. 2. for a visualization of the anticipated results.

In addition to our main hypotheses related to the back matter text's effect on fraction equivalence and fraction arithmetic accuracy, we also explored relations among caregivers' math anxiety, preferences for reading fraction books with their children, and their fraction arithmetic and fraction equivalence performance. Furthermore, we explored whether predictive metacognitive judgments on the fraction arithmetic and fraction equivalence tasks increased from pretest to posttest by experimental condition. Lastly, we explored participants' reports about numeracy activities in the home.

Method

Participants

Based on our preregistered power analysis (https://osf.io/ny5mf?view_only=c944251dea254064acfe481d54335f54 https://osf.io/ny5mf?view_only=3cb31eb0ebfe4d16ab9da69f5f54ee5e), we recruited a total of 160 caregivers aged 18 to 65 years from the United States for this survey on the online platform Prolific. G*Power indicated that we would need 128 participants to detect a small-medium effect size ($f = .15$) for a between-within analysis of variance (ANOVA) with power = 80% and $\alpha = .05$. We oversampled by approximately 20% to account for possible attrition and poor responding and to aim for equal numbers of participants in our four experimental groups. All participants in the current study provided informed consent to participate. Using Prolific's prescreening function, participants were able to complete the study only if they met two inclusion criteria.

The first criterion was that participants have a firstborn child born between the years 2010 and 2020 (i.e., ages about 2 to 12 years). We chose this criterion because picture books are typically marketed to children up through second or third grade (personal communication from Angela Isaacs, children's picture book author), and the Common Core State Standards in Mathematics focus on fraction and ratio standards for children from third grade through sixth grade. The most commonly reported birth year was 2013 (mode: $n = 21$). These data were collected in 2022, meaning that children born in 2013 would have been approximately 9 years old at the time of data collection. The second criterion was that participants' educational attainment be less than or equal to a bachelor's degree. We included this criterion to deter caregivers with exceptionally advanced math knowledge from participating.

We also included several attention checks to ensure that participants were meaningfully engaged in completing the study. One attention check assessed whether participants could identify what visual display (circle or number line) and number type (fractions or negative numbers) they saw in the back matter intervention, and another attention check was included among the questions within the Home Numeracy Scale to assess whether participants were actively engaging with the questions (i.e., select "about once a week"). All participants responded correctly to the attention checks and responded within preregistered criteria for completion time (i.e., no participants needed to be excluded because they completed the survey in less than one third of the average time). The final analytic sample included the full sample of 160 participants. Table 1 includes the demographic characteristics of the sample. A majority of the participants (55%, 88 of 160) held a bachelor's degree and were employed full-time (62.5%, 100 of 160).

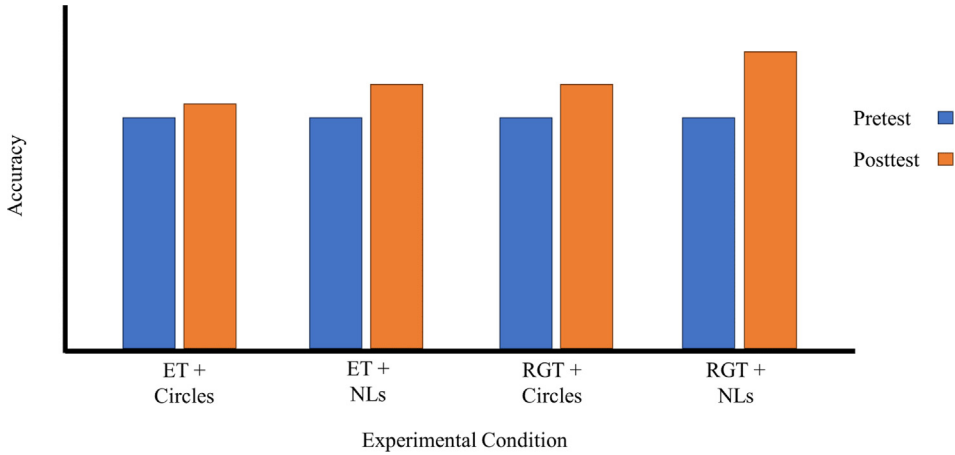


Fig. 2. Anticipated pretest-to-posttest changes in math accuracy across conditions. ET, existing back matter text; NLs, number lines; RGT, researcher-generated back matter text.

Materials

We presented participants with a page of back matter related to fraction concepts via a Qualtrics survey. The back matter included instructions with accompanying visual displays of either circles or number lines (see [Appendix A](#)). Note that participants were not shown any other parts of the books—only the back matter.

At the onset of this project, with the help of a librarian, we searched for a number of children's picture books with fraction content. The primary reason our team conducted this search was to help shape the stimuli for the current study and to get a sense of fraction books currently available to caregivers. Roughly 57% (12 of 21) of the fraction books we analyzed contained back matter, 24% (5 of 21) included circle area models in the back matter, and 10% (2 of 21) included rectangle area models. No books included number lines as part of the back matter, although 3 (14%) included number lines in the book (see [Table S1](#)).

Design and procedure

The current study involved a 2 (Time: pretest vs. posttest, within participants) $\times$ 2 (Back Matter Text Type: ET vs. RGT, between participants) $\times$ 2 (Visual Display Type: circle vs. number line, between participants) between-within design. First, participants completed pretest questions measuring their fraction equivalence accuracy and arithmetic accuracy. Then, participants were assigned to one of four fully crossed experimental conditions. Participants were randomly assigned to a condition in which they saw either instructional back matter text from *Fractions in Disguise* (i.e., ET—unchanged existing back matter text taken directly from *Fractions in Disguise*) or researcher-generated back matter text (RGT). Accompanying the instructional text, participants saw one type of visual display: circles or number lines (see [Fig. 1](#)). Thus, there were four conditions: (a) ET + Circle Visual Displays (Circles), (b) RGT + Circles, (c) ET + Number Line Visual Displays (Number Lines), and (d) RGT + Number Lines. The back matter was divided into two sections, with the first section describing instructions for fraction equivalence and the second section describing instructions for fraction arithmetic.

The main differences between the ET and RGT conditions were (a) ET focused on finding the least common denominator in order to find equivalent fractions with the aim of solving arithmetic problems, whereas RGT focused on equivalent fractions having different whole-number numerator and denominator components; (b) the back matter included explicit use of part-whole language in RGT but not in ET; and (c) an algorithmic focus on arithmetic (ET) compared with a conceptual focus of

Table 1
Participant self-reported characteristics.

Response	<i>n</i>	Percent
Gender		
Female	112	70
Male	48	30
Current employment status		
Employed for wages	100	62.5
Self-employed	16	10
Out of work for > 1 year	7	4.4
Out of work for < 1 year	2	1.3
Homemaker	30	18.8
Student	4	2.5
Did not report	1	0.6
Type of device survey was taken on		
Laptop/desktop computer	137	85.6
Mobile device/cell phone	17	10.6
Other	6	3.5
Race and ethnicity		
White	118	73.8
Black or African American	10	6.3
Asian	9	5.6
Hispanic or Latino	8	5.0
Did not report	1	0.6
Multiple	14	8.8
Highest educational attainment		
High school diploma or GED	14	8.8
Vocational/Trade/Technical school	5	3.1
Some college (no degree)	36	22.5
Associate's degree	17	10.6
Bachelor's degree	88	55
Household income		
\$0 to \$9,999	5	3.1
\$10,000 to \$14,999	2	1.3
\$15,000 to \$19,999	5	3.1
\$20,000 to \$34,999	14	8.8
\$35,000 to \$49,999	25	15.6
\$50,000 to \$74,999	35	21.9
\$75,000 to \$99,999	33	20.6
\$100,000 to \$199,999	31	19.4
\$200,000 or more	8	5
Did not report	2	1.3
Math courses taken	<i>M</i> (<i>SD</i>)	Range
Sum of math courses	4.7 (2.5)	1–12

Note. Several of the demographic questions (e.g., race, gender, employment status) included more inclusive options than none of the participants selected. We present data in this table only if at least 1 participant selected an option.

adding magnitudes together in the RGT condition (see [Appendix A](#) for each condition's back matter). RGT builds on the integrated theory of numerical development ([Siegler et al., 2011](#)) and the fact that the same location on a number line can be represented in an infinite number of ways ([Schiller & Siegler, 2023](#)). Participants were required to spend at least 45 s reading over the first half of the back matter and looking at the visual displays and at least 1 min looking at the second half.³

After reviewing the back matter, participants completed posttest questions measuring fraction equivalence and arithmetic accuracy. All participants were compensated \$4 for completing the approximately 25-min survey ($M = 26.53$ min, $SD = 19.72$). See [Fig. 3](#) for the survey flow and the Open Science Framework (OSF) for exported survey items.

³ Participants could spend as long as they wanted reviewing the back matter; however, most participants spent little more time than the minimum required time (see supplementary material and Table S4 for more information on timing).

Tasks

The tasks in this survey measured participants' objective fraction ability and their subjective attitudes and anxieties about math in general and fractions specifically (see [Tables 2 and 5](#) in Results for descriptive statistics).

Fraction equivalence and predictive metacognitive judgments

Participants completed a set of fraction equivalence problems at pretest and posttest to determine whether their randomly assigned experimental intervention affected fraction equivalence accuracy. We adapted eight true-or-false fraction equivalence problems from [Fitzsimmons et al. \(2020\)](#). We presented participants with one problem at a time in a randomized order. Participants indicated whether the two fractions were equivalent (true) or not (false). Four of the questions were true and four were false (see [Table B1](#) in [Appendix B](#)). Participants were not given feedback on the accuracy of their answers to prevent practice effects. Our dependent variable was percentage correct out of eight items. Participants completed the same eight problems at both pretest and posttest. Immediately before completing the equivalence problems, caregivers indicated how many, out of eight total fraction equivalence problems, they expected to answer correctly. We measured prospective metacognitive judgments because we were interested in exploring whether experiences with picture books affected people's judgments of how well they *expect* they can do on math problems instead of how well they think they *did* on prior problems.

Fraction arithmetic and predictive metacognitive judgments

Participants completed eight fraction arithmetic problems (four addition and four subtraction, four with same denominators and four with different denominators) in a randomized order at pretest and posttest (see [Table B2](#) in [Appendix B](#)). Each problem was presented on its own page, and participants responded by typing the numerator and denominator of their answer into separate boxes. When scoring answers for accuracy, we accepted any fraction that had an equivalent magnitude to the correct answer. Participants were not given feedback on the accuracy of their responses. Our dependent variable was percentage correct out of eight items. Just as with fraction equivalence problems, participants completed the same eight fraction arithmetic problems at pretest and posttest.

Similar to fraction equivalence, participants made a predictive metacognitive judgment about how many total fraction arithmetic problems, out of eight, they expected to answer correctly. In addition to this judgment, participants also made four additional predictive judgments about their expected accuracy: (a) fraction addition problems (out of four), (b) fraction subtraction problems (out of four), (c) fraction arithmetic problems with the same denominator (out of four), and (d) fraction arithmetic problems with different denominators (out of four). Participants answered these additional problems for fraction arithmetic but not for fraction equivalence because the eight fraction arithmetic problems could be easily divided into potentially relevant subcategories (e.g., subtraction is often more challenging than addition; [Siegler et al., 2011](#)), whereas such a distinction was not relevant for the fraction equivalence problems. Note, however, that we determined metacognitive judgments split by subcategories to be outside the scope of the current study, and we do not explore this distinction further in the current article.

Math anxiety

At pretest only,⁴ participants rated how math anxious they felt in different contexts and when working with different types of numbers ([Ashcraft, 2002](#); [Scheibe et al., 2023](#); [Thompson, Taber, et al., 2021](#)). Participants responded to math anxiety questions on two scales: the Formal/Informal Group/Solo (FIGS) scale and the number type math anxiety (NT-MA) scale ([Fitzsimmons et al., 2023](#); [Thompson, Taber, et al., 2021](#)). Both scales assessed anxiety on a 10-point scale with questions beginning with "On a scale

⁴ Due to researcher error, the scales used to measure math anxiety were not included again in the survey after the intervention phase. Because we did not collect posttest math anxiety results, data from the current study cannot be used to evaluate whether any of the back matter conditions influenced math anxiety. However, open questions remain regarding how best to evaluate pretest-to-posttest changes in math anxiety. This point is addressed further in the Discussion.

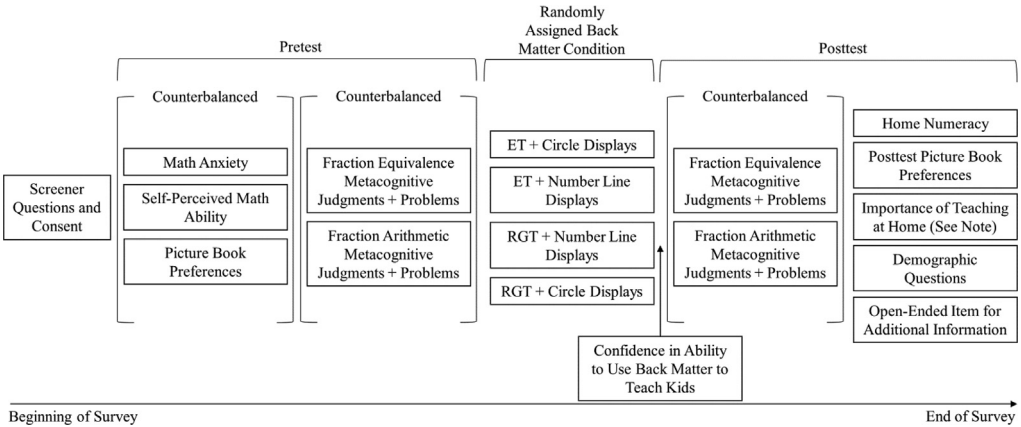


Fig. 3. Survey flow. ET, existing back matter text; RGT, researcher-generated back matter text. We did not include items pertaining to importance of teaching at home in our correlation matrix because caregivers were given the option to say that their children did not attend virtual or in-person school during the pandemic (i.e., If your child did not attend in-person/virtual school between March 2020 and now, please select “not applicable”). However, all caregiver participants reported that their children attended virtual school during that time. Therefore, a substantial number of caregivers said that the questions about teaching math at home were not applicable to them.

Table 2

Means and standard deviations for fraction equivalence and arithmetic accuracy at pretest and posttest.

	Pretest equivalence		Posttest equivalence		Pretest arithmetic		Posttest arithmetic	
	M	SD	M	SD	M	SD	M	SD
ET + circle visual displays (<i>n</i> = 42)	77.1%	17.7%	77.7%	18.0%	59.8%	35.1%	71.4%	30.2%
RGT + circle visual displays (<i>n</i> = 40)	73.4%	18.4%	79.7%	17.1%	73.8%	28.1%	79.4%	27.2%
ET + number line visual displays (<i>n</i> = 38)	78.6%	15.9%	82.6%	15.5%	75.3%	31.6%	79.6%	31.0%
RGT + number line visual displays (<i>n</i> = 40)	76.6%	17.7%	76.3%	17.2%	63.4%	33.9%	71.3%	28.1%

Note. ET, existing back matter text; RGT, researcher-generated back matter text.

of 1 to 10, with 10 being the most anxious, how math anxious ...” The FIGS scale is a 16-item scale that measures math anxiety in four different contexts: (a) formal solo contexts (e.g., “... would you feel about completing an unannounced pop quiz in a math class?”), (b) formal social contexts (e.g., “... would you feel about being given a set of problems to solve on the board in front of your classmates?”), (c) informal solo contexts (e.g., “... would you feel about doubling a recipe when cooking?”), and (d) informal social contexts (e.g., “... would you feel about figuring out an appropriate tip on a shared bill at a restaurant?”). Scale reliability for the FIGS scale was strong in the current sample ($\alpha = .93$). The NT-MA scale is a five-item scale that measures math anxiety in general as well as for different number types (i.e., whole numbers, fractions, whole-number frequencies, and percentages; Thompson, Taber, et al., 2021). Scale reliability for the NT-MA scale was moderately strong in the current sample ($\alpha = .86$). Due to a coding error, these math anxiety measures were administered only at pretest instead of at both pretest and posttest as we had intended.

Self-perceived math ability

Participants indicated how good they felt they were at math in general and with fractions (from the Math Attitudes Questionnaire [MAQ]; Mielicki et al., 2022; Sidney et al., 2021). Participants rated their math ability (i.e., “How good are you at math?”) on a 6-point Likert-like scale ranging from *very bad* to *very good*. We treated this seven-item measure as exploratory in our analyses. Scale reliability for the MAQ subscale was strong in the current sample ($\alpha = .91$).

Picture book preferences

At pretest and posttest, participants used a sliding scale from 0% (*not at all likely*) to 100% (*completely likely*) to rate themselves on the following question: “How likely are you to read a book that includes information about math, like fractions, to your child?” The aim of this was to explore whether exposure to our back matter fraction instruction intervention affected participants’ desires to read books containing math concepts to their children.

Confidence in using the presented back matter to teach participants’ children

After the intervention, participants placed their rating on a sliding scale from 0% (*I’m not confident at all*) to 100% (*I’m completely confident*) for the following question: “How confident are you in your ability to use what you learned in the fraction instruction to teach your child about fractions?” Caregivers’ metacognitive judgments regarding their ability to use the back matter to help teach their children may be important to consider because caregivers might not use back matter if they do not feel confident in doing so. Prior research has shown that metacognitive judgments about math are related to math anxiety (Morsanyi et al., 2019; Scheibe et al., 2022), math self-concept (Ahmed et al., 2012), math self-efficacy (Pajares & Miller, 1994), and math attitudes (Scheibe et al., 2022). All these factors may interact to affect how caregivers use back matter to help their children. We treated this metacognitive measure as exploratory in our analyses.

Home numeracy

Finally, participants rated how often they engaged in direct and indirect math talk and practice in their home. We adapted the Home Numeracy Scale (Thompson et al., 2017) to include additional activities that might be more relevant for older children with advanced fraction knowledge. Participants rated their frequency of 18 direct and indirect math activities on a 6-point Likert-like scale from *never* to *multiple times per day*. The direct activities included counting objects or people, writing down numbers, reading number storybooks, using number activity books, using the term “more”/“less”, counting down (10, 9, 8, 7, ...), learning math facts, identifying names of written numbers, using fraction words like “one half,” “one third,” and “one fourth”, playing with number/math apps on a tablet, phones, or computer, and using a ruler or measuring tape. The indirect activities included using calendars and dates, measuring ingredients when cooking, playing board games with a die or spinner, engaging in “connect-the-dot” activities, counting out money, making/sorting collections, playing “store”, playing card games, sorting things by size, color, or shape, talking about money when shopping (“which costs more?”), and creating or identifying patterns. We treated this measure as exploratory in our analyses. Scale reliability for the Home Numeracy Scale was strong in the current sample ($\alpha = .92$).

Results

Analytic plan overview

First, we report checks of random assignment of participants to the four fully crossed experimental conditions. Then, we report our preregistered mixed ANOVAs assessing pretest-to-posttest changes in fraction equivalence and fraction arithmetic performance. We next report correlations among math anxiety and other tasks. Finally, we end our Results section with exploratory analyses pertaining to (a) pretest-to-posttest changes in predictive metacognitive judgments on equivalence and arithmetic performance and (b) whether caregivers’ picture book preferences or confidence in conveying to their children the information that was presented by the back matter differed depending on which of the four back matter conditions the caregivers engaged with.

Random assignment

We conducted separate 2 (ET vs. RGT) $\times 2$ (circle displays vs. number line displays) between-participants ANOVAs (Table 2) on fraction equivalence accuracy ($p = .774$), fraction arithmetic accu-

racy ($p = .013$), predictive metacognitive judgments ($p = .462$), self-perceived math ability ($p = .501$), and math anxiety (NT-MA: $p = .154$; FIGS: $p = .023$) at pretest. At pretest, participants were more accurate in the fraction arithmetic task in the ET + Number Lines condition than in the ET + Circles condition ($p = .034$). In addition, participants in the RGT + Number Lines condition reported significantly lower math anxiety on the FIGS math anxiety scale compared with participants in the RGT + Circles ($p = .028$) and ET + Number Lines ($p = .031$) conditions. That is, based on pairwise comparisons, random assignment was unsuccessful for fraction arithmetic and for the FIGS measure of math anxiety specifically. We explored the FIGS and NT-MA math anxiety scales separately as covariates in our models because the two scales operationalize math anxiety differently, and the FIGS scale is not yet fully validated. We decided to maintain our preregistered between-within ANOVA examining changes over time in our dependent variables of interest despite these noted pretest differences across conditions.⁵

Fraction accuracy as a function of experimental condition and time

We hypothesized that there would be a three-way interaction among timing (i.e., pretest vs. posttest), the type of back matter text (i.e., ET vs. RGT), and the type of visual display (i.e., circles vs. number lines) on both the fraction equivalence and fraction arithmetic tasks. We expected participants who were randomly assigned to the RGT + Number Lines condition to show the greatest improvements in their fraction equivalence and arithmetic scores from pretest to posttest in comparison with participants in the other three conditions.

Fraction equivalence

In a 2 (Back Matter Text Type: ET vs. RGT, between participants) $\times$ 2 (Visual Display: number lines vs. circles, between participants) $\times$ 2 (Time: pretest vs. posttest, within participants) between-within ANOVA (see Table 3), there was a significant three-way Time $\times$ Visual Display $\times$ Back Matter Text Type interaction ($p = .041$) (see Fig. 4). Planned pairwise comparisons indicated that equivalence accuracy increased from pretest to posttest in the RGT + Circles condition only ($p = .010$). None of the two-way interactions was significant, and the only significant main effect was for time (i.e., improvement from pretest to posttest, $p = .031$). See Table 3 for model details.

Fraction arithmetic

There was no significant three-way Time $\times$ Visual Display $\times$ Back Matter Text Type interaction on arithmetic accuracy (see Fig. 5). However, there was a significant Visual Display $\times$ Back Matter Text Type two-way interaction, $F(1, 156) = 5.004$, $p = .026$ (see Table 4 for model details). Follow-up analyses of the two-way interaction indicated that, collapsed across pretest and posttest, participants were more accurate in the RGT + Circles ($M = 76.56\%$, $SD = 26.73$) and ET + Number Lines ($M = 77.47\%$, $SD = 30.19$) conditions relative to the RGT + Number Lines ($M = 67.34\%$, $SD = 29.91$) and ET + Circles ($M = 66.63\%$, $SD = 31.49$) conditions (see Fig. 5). Although there was not a significant three-way interaction, planned pairwise comparisons indicated that arithmetic accuracy increased from pretest to posttest in the ET + Circles ($p < .001$), RGT + Circles ($p = .030$), and RGT + Number Lines ($p = .003$) conditions. None of the other two-way interactions was significant, and the only significant main effect was for time ($p < .001$).

Correlations among tasks

We explored correlations between self-perceived math ability (from the MAQ) and accuracy on fraction equivalence and arithmetic problems. As seen in Table 5, self-perceived math ability was positively correlated with pretest ($r = .44$, $p < .01$) and posttest ($r = .33$, $p < .01$) fraction equivalence accuracy and with pretest ($r = .44$, $p < .01$) and posttest ($r = .38$, $p < .01$) fraction arithmetic accuracy. That

⁵ Because of pretest differences on the FIGS scale, we also ran our models with the FIGS scale included as a covariate. There were no significant changes when the scale was included as a covariate; thus, we do not report these statistics.

Table 3

Fraction equivalence model details.

	<i>F</i>	<i>p</i>	η_p^2
Time ^a	4.73	.031	.029
Visual display	0.39	.534	.002
Back matter text type	1.05	.308	.007
Visual Display × Back Matter Text Type	0.47	.493	.003
Time × Visual Display	0.45	.506	.003
Time × Back Matter Text Type	0.08	.772	.001
Time × Visual Display × Back Matter Text Type	4.24	.041	.026

Note. Degrees of freedom for all *F* tests were (1, 156).

^a Time represents pretest-to-posttest change.

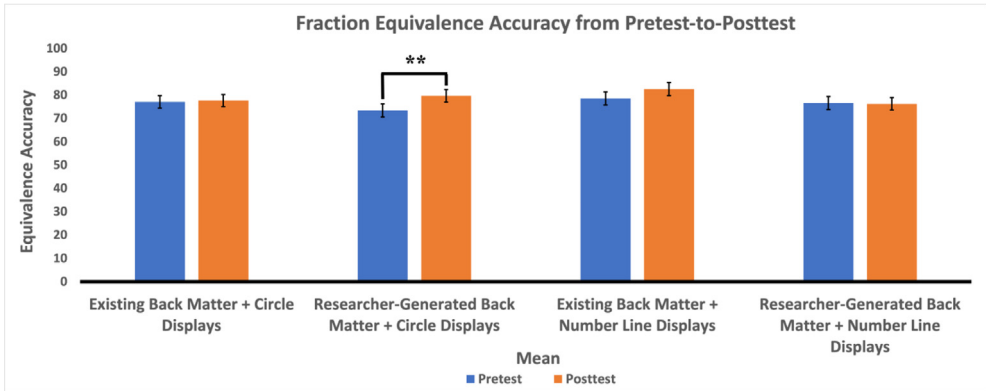


Fig. 4. Fraction equivalence accuracy from pretest to posttest. Error bars represent standard error. **Significant at the .01 level.

is, we found that participants who had higher self-perceived math ability were more accurate on the fraction equivalence and arithmetic tasks.

Correlations with math anxiety

In addition to correlations between self-perceived math ability and math performance, we explored relations among math anxiety and math performance. Math anxiety on the five-item NT-MA scale was negatively correlated with pretest ($r = -.45, p < .01$) and posttest ($r = -.39, p < .01$) fraction equivalence accuracy and with pretest ($r = -.39, p < .01$) and posttest ($r = -.36, p < .01$) fraction arithmetic accuracy. That is, lower math anxiety was related to higher performance on both fraction tasks. Math anxiety on the 16-item FIGS scale was also negatively correlated with pretest ($r = -.37, p < .01$) and posttest ($r = -.33, p < .01$) fraction equivalence accuracy and with pretest ($r = -.27, p < .01$) and posttest ($r = -.26, p < .01$) fraction arithmetic accuracy. Results revealed the two math anxiety measures to be strongly correlated ($r = .78, p < .01$). Thus, participants' responses on both math anxiety measures were consistent with prior literature on the reliable negative relation between math anxiety and math performance (e.g., Hembree, 1990).

Exploratory analyses

In addition to the primary preregistered models and relations between study variables, we also explored several analyses. The focus for the remainder of the Results section is on whether the experimental manipulation affected participants' metacognitive judgments: (a) their predictive metacognitive judgments, (b) their confidence in ability to use back matter, and (c) their reported preferences for using picture books.

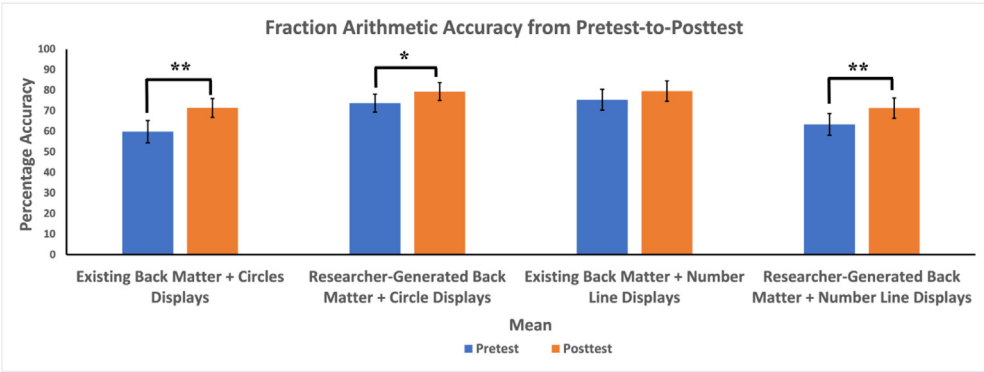


Fig. 5. Fraction arithmetic accuracy from pretest to posttest. Error bars represent standard error. *Significant at the .05 level. **Significant at the .01 level.

Table 4
Fraction arithmetic model details.

Variable	<i>F</i>	<i>p</i>	η_p^2
Time ^a	31.45	<.001	.168
Visual display	0.08	.780	.001
Back matter text type	0.01	.931	.000
Visual Display × Back Matter Text Type	5.04	.026	.031
Time × Visual Display	0.97	.327	.006
Time × Back Matter Text Type	0.22	.641	.001
Time × Visual Display × Back Matter Text Type	3.31	.071	.021

Note. Degrees of freedom for all *F* tests were (1, 156).

^a Time represents pretest-to-posttest change.

Predictive metacognitive judgments

We conducted an exploratory 2 (Back Matter Text Type: ET vs. RGT, between participants) × 2 (Visual Display: number lines vs. circles, between participants) × 2 (Time: pretest vs. posttest, within participants) between-within ANOVA on predictive metacognitive judgments on both the fraction arithmetic task and fraction equivalence task. The specific judgment of interest prompted participants to provide a predictive judgment about how many fraction arithmetic problems or fraction equivalence problems out of eight they thought they would answer correctly. For fraction arithmetic accuracy, a main effect of time, $F(1, 156) = 4.55$, $p = .034$, $\eta_p^2 = .03$, indicated that participants expected to do better at posttest ($M = 4.65$, $SD = 2.51$) relative to pretest ($M = 4.36$, $SD = 2.30$). That is, after seeing the back matter, participants—collapsed across all four experimental conditions—predicted that they would perform better at posttest than at pretest for fraction arithmetic specifically. However, there were no other main effects or interactions for fraction arithmetic and no main effects or interactions for fraction equivalence (all $ps > .20$), and the main effect of time for fraction arithmetic was a small effect ($\eta_p^2 = .03$).

Confidence in ability to use back matter

Participants' predictive metacognitive judgments regarding fraction equivalence and fraction arithmetic were specific to the stimuli within the current study. However, caregivers also made broader metacognitive judgments regarding their perceptions of back matter. For example, immediately after participants engaged with the back matter intervention, we asked them to rate their confidence on a slider scale (0% = *I'm not confident at all* to 100% = *I'm totally confident*) in their ability to use the back matter they just saw to teach their children about fractions. Those who endorsed higher confidence in using the back matter to teach their children about fractions were also more accurate at pretest

Table 5
Descriptive statistics and correlations for study variables.

	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7	8	9	10
1. Pretest fraction equivalence accuracy ^a	76.4%	17.4%	$\alpha = .48$									
2. Posttest fraction equivalence accuracy ^a	79.0%	17.0%	.61**	$\alpha = .44$								
3. Pretest fraction arithmetic accuracy ^a	67.9%	32.7%	.52**	.47**	$\alpha = .89$							
4. Posttest fraction arithmetic accuracy ^a	75.3%	29.1%	.54**	.48**	.86**	$\alpha = .87$						
5. FIGS math anxiety scale ^b	4.69	1.86	-.37**	-.33**	-.27**	-.26**	$\alpha = .93$					
6. NT-MA math anxiety scale ^b	3.95	1.95	-.45**	-.39**	-.39**	-.36**	.78**	$\alpha = .86$				
7. Self-perceived math ability ^b	4.21	1.00	.44**	.33**	.43**	.38**	-.68**	-.77**	$\alpha = .91$			
8. Home Numeracy Scale ^b	2.10	0.81	.07	.02	.13	.11	-.16*	-.08	.16*	$\alpha = .92$		
9. Pretest picture book preference ^c	60.23	30.78	.10	.16*	.11	.07	-.25**	-.25**	.25**	.31**		
10. Posttest picture book preference ^c	60.79	29.95	.24**	.24**	.24**	.19*	-.32**	-.34**	.33**	.43**	.75**	
11. Confidence in using back matter to teach children ^d	58.32	31.87	.47**	.44**	.51**	.49**	-.43**	-.54**	.57**	.15	.19*	.41**

Note. The *n* for all variables in Table 5 was 160. Scale reliability (Cronbach's alpha) is reported on the diagonal. The low scale reliability for fraction equivalence may have been due to the small number of two-choice items and variability in equivalence items, with some items being more challenging than other items, leading to more variability in the responses. No reliability is reported for picture book preferences or confidence using back matter to teach children because caregivers responded to one item for each construct.

**p* < .05.

***p* < .01.

^a Fraction equivalence and arithmetic accuracy (numbers 1–4) were scored on a 0% (completely incorrect) to 100% (completely correct) scale.

^b Self-report measures of math anxiety, math attitudes, and home numeracy (numbers 5–8) were scored on the varying parameters of each scale: Formal/Informal Group/Solo (FIGS)—1 (*not anxious*) to 10 (*very anxious*); number type math anxiety (NT-MA)—1 (*not anxious*) to 10 (*very anxious*); self-perceived math ability—1 (*very bad*) to 6 (*very good*); Home Numeracy Scale—1 (*never*) to 6 (*multiple times per day*).

^c Caregivers' preferences for reading a math book to their children (numbers 9 and 10) was scored on a slider scale from 0 (*not at all likely*) to 100 (*completely likely*).

^d Caregivers' confidence in their ability to use what they learned to teach their children about fractions (number 11) was scored on a slider scale from 0 (*I'm not confident at all*) to 100 (*I'm totally confident*).

($r = .47, p < .01$) and posttest ($r = .44, p < .01$) on the fraction equivalence task. Caregivers' confidence in using back matter was also moderately positively related to their accuracy on the pretest ($r = .51, p < .01$) and posttest ($r = .49, p < .01$) on the fraction arithmetic tasks. However, there were no differences in the magnitude of participants' confidence judgments for using back matter by intervention groups, $F(1, 159) = 0.29, p = .589$. Thus, participants' confidence in using back matter beyond the current study was moderately related to their accuracy on math stimuli within the study, but this relation was not specific to particular experimental conditions.

Picture book preferences and home numeracy

To stick with the theme of caregivers' judgments about picture books, we explored whether there was an effect of back matter type, visual display type, time, or any two- and three-way interactions on parents' preferences for reading their children a fraction picture book in a 2 (Back Matter Text Type: ET vs. RGT, between participants) $\times$ 2 (Visual Display: number lines vs. circles, between participants) $\times$ 2 (Time: pretest vs. posttest, within participants) between-within ANOVA. Results yielded no significant three-way (or two-way) interactions (see Table S3 in [supplementary material](#) for more details). However, participants' math anxiety was negatively related to their endorsement of reading a fraction picture book (NT-MA: $r = -.34, p < .01$; FIGS: $r = -.32, p < .01$). Thus, it appears that the current study provides some evidence that participants with higher levels of reported math anxiety may be less likely to read fraction picture books to their children.

Finally, we explored participants' ratings on an adapted version of the Home Numeracy Scale. The scale was not statistically related to equivalence performance or arithmetic performance at pretest or posttest. However, results from a between-within ANOVA revealed a significant three-way interaction for type of home numeracy activity (i.e., direct vs. indirect activities), back matter text type, and visual display type, $F(1, 156) = 5.58, p = .019, \eta_p^2 = .04$ (see [supplementary material](#) for more details and descriptive statistics).

Discussion

In this study, caregivers were randomly assigned to one of four conditions in a fully crossed 2 (Back Matter Text Type: ET vs. RGT, between participants) $\times$ 2 (Visual Display: number lines vs. circles, between participants) pretest–posttest experimental design. The goal of the back matter was to explain to caregivers how to solve fraction equivalence and arithmetic problems. We compared existing back matter text (ET) with researcher-generated back matter text (RGT), informed by research in mathematical cognition. The goal of this comparison was to assess whether providing clear definitions of fraction terminology and illustrating step by step how parts and wholes are related to one another might facilitate caregivers' fraction learning and to test whether the back matter text interacted with the visual display used to represent fractions.

Our first preregistered hypothesis was that researcher-generated back matter text (RGT) would improve participants' posttest math performance more than existing back matter text (ET). That is, we expected participants who saw RGT—regardless of the visual display that was paired with the text—to show greater improvements from pretest to posttest on fraction equivalence, as well as on fraction arithmetic, relative to participants who saw ET. In contrast to our expectations, results indicated no main effect of back matter text type for either fraction equivalence performance (see Fig. 4 and Table 3) or fraction arithmetic performance (see Fig. 5 and Table 4). Rather, planned comparisons indicated that caregivers' equivalence accuracy significantly increased from pretest to posttest only in the RGT + Circles condition (see Fig. 4). Fraction arithmetic accuracy revealed different findings than fraction equivalence. Fraction arithmetic accuracy increased from pretest to posttest in three of the four conditions: ET + Circles, RGT + Circles, and RGT + Number Lines.

Our second preregistered hypothesis was that participants who saw the RGT paired with the number line visual display would show the largest improvements in fraction equivalence and fraction arithmetic accuracy from pretest to posttest after engaging with the back matter. Again, contrary to our hypothesis, the results did not support the proposed three-way interaction for either fraction equivalence performance (see Fig. 4) or fraction arithmetic performance (see Fig. 5). Results indicated

a significant three-way interaction of Time $\times$ Visual Display $\times$ Back Matter Text Type, but not in the way we predicted—only a significant increase in participants' accuracy in the RGT + Circles condition from pretest to posttest (see Fig. 4 and Table 3). Results indicated no significant three-way interaction for fraction arithmetic (see Fig. 5 and Table 4). To summarize our primary results, the four fully crossed back matter conditions did seem to affect posttest math performance outcomes, yet not in the ways we anticipated they would. We explore possible explanations for the results throughout the rest of the Discussion.

Back matter affects math outcomes

Overall, participants' pretest-to-posttest improvements in fraction arithmetic performance in three of the four conditions adds to the literature about the benefits of interventions when teaching fractions (Fuchs et al., 2016) and extends this work to caregivers. Results suggest that caregivers found familiar circle visual displays helpful when solving fraction arithmetic problems regardless of the type of back matter text. However, caregivers benefited from number line displays only when they were accompanied by researcher-generated instructions. Thus, caregivers likely benefit from visual displays—both circles and number lines—although perhaps for different reasons. Note, however, that our mixed findings between fraction arithmetic and fraction equivalence highlight that math as a domain involves a variety of different tasks that require different skills. Thus, future research should investigate under what circumstances and for what specific outcomes back matter may be most effective at aiding caregivers. Open questions remain about how best to help caregivers learn from back matter by incorporating evidence from math cognition on the efficacy of (a) visual displays in helping learners to activate relevant prior knowledge (e.g., Sidney et al., 2022) and (b) number lines in facilitating relational reasoning (Thompson, Taber, et al., 2021).

Importantly, these results add to a growing body of literature (e.g., Fitzsimmons et al., 2023; Thompson, Taber, et al., 2021) suggesting that even very brief online interventions have the potential to improve adults' understanding of rational numbers. Indeed, participants interacted with the back matter for only 2 to 3 min on average (see Table S4 in supplementary material), yet many participants still demonstrated significant posttest gains in fraction arithmetic accuracy and some participants also demonstrated significant posttest gains in fraction equivalence accuracy. Thus, the current study provides an encouraging contribution regarding the efficacy of light-touch interventions that focus on worked examples, which provide step-by-step instructions on how to solve problems.

Compelling evidence exists for a consistent moderate effect of worked examples improving math outcomes (Barbieri et al., 2023), particularly for learning in math-related domains (Booth et al., 2015). Research on worked examples suggests that these tools are effective in part because they reduce problem-solvers' cognitive load (Sweller, 1988; see also Barbieri et al., 2023). That is, step-by-step procedures simplify the problem-solving process, thereby reducing the need for a heavy reliance on working memory resources (e.g., trying to remember the algorithms to find equivalent fractions or complete a fraction arithmetic problem). Reducing the reliance on working memory resources may be particularly important for caregivers when reading to their children because processing the book content is only one component of the task: They likely need working memory resources to monitor how their children are comprehending and enjoying the story. Worked examples presented through back matter may be an ideal tool to ease this process, especially if caregivers refer to the back matter as they read the book to their children.

The current study extends prior research on worked examples and the efficacy of math-related picture books by promoting research-informed worked examples through back matter in children's fraction books. Given recent promising findings suggesting that caregivers' reading of math-related picture books to their children likely promotes math outcomes (Purpura et al., 2021; Uscianowski et al., 2020), combining worked examples informed by research in math cognition with children's fraction picture books is a natural extension. In addition, prior research has been mainly conducted with simpler math content and younger children. The current work is a novel extension that builds on the prior work with young children and simple math vocabulary (e.g., Purpura et al., 2021; Uscianowski et al., 2020). Thus, future research should continue to build on prior research and the current findings

to continue improving on the tools caregivers have to aid their children's math learning in the home math environment (Hornburg et al., 2021).

Back matter and home numeracy

Not only do back matter features affect math outcomes for caregivers, but initial evidence suggests that manipulations to back matter may also affect a variety of caregivers' metacognitive judgments. In addition, caregivers' metacognitive judgments are related to math anxiety, and math anxiety is negatively related to math performance (e.g., Hembree, 1990). Thus, the current data provide evidence for relations between back matter, metacognitive judgments and math anxiety. In this section, we recap some exploratory findings from the current study, briefly tie in the metacognition and math anxiety literature, and extrapolate on current findings to speculate about important interactions between these variables and how they may be related to the home numeracy environment.

Metacognition—broadly construed—is typically considered thinking about thinking (Flavell, 1979). Metacognition as a construct was relevant to the current study for two primary reasons, namely that (a) metacognitive experiences are a component of problem-solving and (b) caregivers made metacognitive judgments about their confidence in using back matter to help their children learn. Regarding metacognition during problem-solving, caregivers in the current study made predictive global judgments about how many fraction arithmetic and fraction equivalence problems they would answer correctly for the respective math tasks. In general, participants predicted they would answer more problems correctly at posttest than at pretest, but results indicated no difference based on experimental condition. Thus, participants' increase in confidence at posttest—relative to pretest—likely had more to do with an increase in knowledge of the task requirements and their sense of how they did on the pretest problems than it did with the type of back matter they viewed. Recent research proposed that metacognitive thoughts and feelings (i.e., metacognitive experiences; Efklides, 2006) and math anxiety (and working memory) are related in predicting problem-solving performance (Morsanyi et al., 2019; Scheibe et al., 2023). However, testing these relations was not the focus of the current study; thus, the current research design does not explicitly clarify these relations. Perhaps more interesting than metacognition in problem-solving is metacognition related to caregivers' confidence in their ability to use back matter to help their children learn. Open questions remain for future research about the possible effects of back matter type to influence caregivers' confidence not only in the moment but also longitudinally in the home environment.

Similar to how caregivers' confidence may affect learning outcomes, so too might their math anxiety. Participants in each condition who rated themselves, at pretest, as more math anxious in a number of different contexts (using the FIGS scale) and for different number types (NT-MA scale) had lower accuracy on fraction equivalence and fraction arithmetic measures at both pretest and posttest. We also analyzed the associations among fraction equivalence performance, fraction arithmetic performance, math anxiety, and caregivers' self-reported frequency of home numeracy practices (see Table 5). The average participant engaged in activities on the Home Numeracy Scale (Thompson et al., 2017) an average of only about once per week ($M = 2.10$, where 0 = *never* and 5 = *multiple times per day*; see Table 5). We also found that caregivers who reported lower levels of math anxiety and higher self-perceived math abilities also tended to report engaging in more math activities with children in the home. Thus, participants' evaluations of their math abilities and math anxiety are associated with their reported behaviors concerning working with numerical concepts with their children in the home.

In a related exploratory analysis, we examined how confident participants were in using what they learned from the back matter to teach their children fraction concepts. The results showed that caregivers who performed more accurately on the fraction tasks gave higher ratings on their self-perceived ability to disseminate that information to their children. This confidence measure was also negatively correlated with both math anxiety scales (FIGS and NT-MA). Evidence from prior studies suggests that math anxiety is related to online, in-the-moment metacognitive judgments and feelings (Bellon et al., 2021; Desender & Sasanguie, 2022; Scheibe et al., 2023), math self-concept (Ahmed et al., 2012; Lee, 2009; Jameson, 2014; Rolison et al., 2020), math self-efficacy (Bandura, 1986; Pajares & Miller, 1994), math motivation (Özcan & Gümüş, 2019; Skaalvik et al., 2015; Zimmerman, 2008), and working memory (Ashcraft & Kirk, 2001; Beilock & Carr, 2005; Passolunghi et al., 2019). It is an open question as to

how these constructs interact to affect caregivers' willingness to convey math information to their children. Future researchers may want to further investigate the interactions among these measures to increase caregivers' fraction equivalence and arithmetic accuracy as well as their confidence in conveying math information to their children. This is especially important given that caregivers' math anxiety can negatively influence their children's math performance and math anxiety (Brown et al., 2023; Maloney et al., 2015).

Limitations and future directions

There are a few potential limitations of the current study that should be noted. One potential limitation that should be considered in the future is that the survey was delivered using an online platform without a researcher observing to ensure that participants were actively engaging with the back matter. It is possible that some participants did not read the back matter carefully—or at all—which could have made the back matter less effective for these individual participants. However, it is important to note that all caregivers attended to the back matter closely enough to correctly answer both attention check questions and spent at least 1 min and 45 s on the back matter pages—and usually more time (see Table S4). To briefly summarize the intervention timing, most participants spent roughly 2 min in reviewing the back matter. Despite such a brief intervention, current results indicated differences for both fraction equivalence and fraction arithmetic by condition. We argue that the brevity of the intervention, paired with the resulting condition differences, is a strength of the current study. Indeed, the current study employed a much more light-touch intervention compared with other recent research with much longer or multi-session interventions with picture books (e.g., Gibson et al., 2020; Purpura et al., 2021, 2024). Regardless of these promising findings, future research should employ mixed methods including close observations of caregivers and children interacting together with fraction picture books. Another consideration regarding the method and sample is that our sample exclusively consisted of caregivers, rather than both caregivers and children, so these data cannot support any conclusions about how caregiver outcomes may influence children's outcomes. However, this study is an important first step toward better understanding how to improve caregivers' fraction understanding, which may be used to facilitate positive math interactions between caregivers and their children in the home learning environment in future work. It is noteworthy that our brief intervention improved caregivers' fraction performance in some conditions because it had likely been some time since our participants initially learned about fractions in formal learning environments.

Despite the fact that caregivers' fraction performance improved in some conditions, the results did not perfectly mirror our predictions. One possible explanation for why we did not see improvements in fraction equivalence performance from pretest to posttest across the other experimental conditions is because the denominators were too large to implement the part-whole strategy of dividing up the number line into equal-sized denominator pieces, a featured strategy presented in the back matter. It is possible that had we given caregivers a fraction (e.g., $15/30$) and asked them to select the visual display that showed an equivalent magnitude (rather than asking them whether an equation was true), they may have been better able to showcase their equivalence knowledge in the way they had just learned about it via the back matter. In addition, the specific task features may have led to results for fraction equivalence and fraction arithmetic not mirroring one another. For example, participants evaluated fraction equivalence with true or false statements (see Table B1 in Appendix B), whereas they answered fraction arithmetic problems by providing answers in open response text boxes (see Table B2 in Appendix B). It is possible that the differences in how participants responded to the two math tasks may have led to the different pattern of results seen in Figs. 4 and 5.

Indeed, participants' responses indicated more variability for fraction arithmetic accuracy (standard deviations of 27.2%–35.1%; see Table 2) than for fraction equivalence accuracy (standard deviations of 15.5%–18.4%; see Table 2). Even though both these tasks rely on a conceptual understanding of fraction magnitudes, the differences between the tasks may have led to differences in performance. It is also possible that the participants did learn conceptually about fraction equivalence, yet if the task demands do not align closely with the type of learning or worked example, transfer challenges (Barnett & Ceci, 2002; Klahr & Chen, 2011) may have played a role in our findings. In addition, as mentioned in our original description of our predictions, adult participants may have learned about

rational numbers primarily using circle visual displays. Thus, even though number lines facilitate rational number understanding (Fitzsimmons et al., 2023; Mielicki et al., 2022; Thompson, Taber, et al., 2021; Yu et al., 2022), the current study may have represented a conflict between the benefit of the number line and the benefit of familiarity with circle visual displays.

Another possible explanation for our findings is that some of the significant differences displayed in Figs. 4 and 5 may reflect regression to the mean (Nisbett, 2015). That is, the only significant increase in fraction equivalence performance from pretest to posttest was in the RGT + Circles condition, and participants in this condition had the lowest pretest fraction equivalence scores. Participants in the RGT + Circles condition averaged 73.4% for their pretest fraction equivalence accuracy, whereas participants in the other three experimental conditions averaged 76.6% to 78.6% (see Table 2). Following this same logic, the largest pretest-to-posttest changes for fraction arithmetic were in the two conditions with the lowest pretest fraction arithmetic accuracy: ET + Circles (59.8% at pretest) and RGT + Number Lines (63.4% at pretest) (see Table 2). Thus, replication is needed to provide more evidence regarding how caregivers interact with back matter. More broadly, open questions remain regarding how parents use back matter naturalistically when interacting with their children in their home and how the effects of teaching their children in the home math environment may differ from experimental manipulations in an online context.

Finally, future research should also investigate the role of other individual differences in the relations among math anxiety, back matter, and metacognition (e.g., confidence in caregivers' math skills). For example, previous research has indicated that even when math skills are equated across men and women, women tend to be less confident than men about their math performance (Rivers et al., 2020). In addition, research suggests that when female elementary math teachers report high math anxiety, this anxiety can negatively affect their students' math performance, possibly via young girls' beliefs about who is good at math (Beilock et al., 2010). A large field study involving an 89% female sample of parents (Maloney et al., 2015) provided supplementary data suggesting that parents' math anxiety is often closely related to children's math achievement and children's math attitudes. The primarily female sample in Maloney and colleagues' study is a common concern for potential sampling bias: Fathers are often underrepresented in research on the home learning environment (Costigan & Cox, 2001). However, it may be that mothers are often spending more time than fathers in teaching their children at home (McKee et al., 2021). It is an open question for future research whether individual differences in confidence by gender exist as caregivers engage with back matter as they read math picture books to their children. There may be important differences based on gender that are not typically revealed in developmental research with parents due to a disproportionate study participation rate of women compared with men.

Conclusions

Understanding fractions is crucial for children's mathematical development, and caregivers play a significant role in children's acquisition of those concepts outside of formal learning environments (Hornburg et al., 2021). The use of back matter in children's fraction picture books can be an effective way for caregivers to review fraction equivalence and fraction arithmetic concepts themselves so that they can help their children learn about fractions. The current study extends our understanding of how different types of back matter (e.g., incorporating number line visual displays or circle visual displays) might affect important caregiver outcomes such as understanding of fraction equivalence and fraction arithmetic. The current study serves as a platform for future research on strategies for creating effective back matter and informs how back matter may relate to other caregiver outcomes such as math anxiety. The positive effects of the back matter were strongest for caregivers who reported feeling less anxiety toward math. This highlights an ever-present need to help adults reduce their math anxiety so that they can support their children in learning fraction concepts and help them to develop positive feelings toward fractions.

Data availability

This study was preregistered on the Open Science Framework (OSF) (https://osf.io/ny5mf?view_only=3cb31eb0ebfe4d16ab9da69f5f54ee5e). All statistical output is available through the OSF.

Code availability

Syntax is available upon request. Output is available through the OSF.

CRediT authorship contribution statement

Daniel A. Scheibe: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Formal analysis, Data curation, Conceptualization. **Lauren Wyatt:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Charles J. Fitzsimmons:** Writing – review & editing, Writing – original draft, Supervision, Formal analysis, Conceptualization. **Marta K. Mielicki:** Writing – review & editing, Writing – original draft, Conceptualization. **Lauren K. Schiller:** Writing – review & editing, Writing – original draft, Conceptualization. **Clarissa A. Thompson:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation.

Acknowledgments

This research was supported by the U.S. Department of Education, Institute of Education Sciences (R305U200004), and by an internal grant from the University Research Council of Kent State.

This manuscript was based on an original honor's thesis project by Lauren Wyatt. Although Dan Scheibe took over as lead on the project, the team would like to acknowledge Lauren's original work and consider Dan and Lauren as co-first authors.

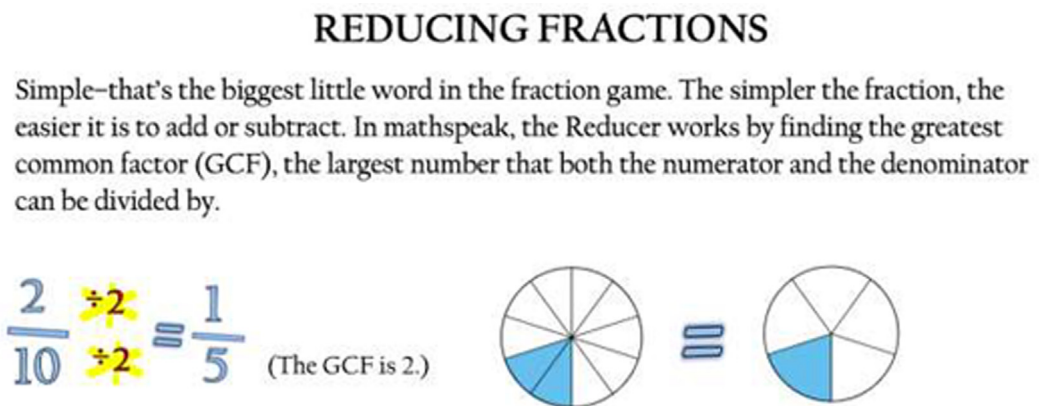
The team would also like to thank Dr. Timothy Scheibe and Angela Scheibe for volunteering their time and assistance with editing during the review process.

Appendix A. Back matter examples

Prior to seeing the back matter, participants in all four conditions read the following instructions: "Now, we're going to show you some instructions on how to find equivalent fractions and add/subtract fractions. Information like this might be found at the back of a children's book. It can aid parents and guardians in helping their children think about the math in the book.

"Please take the time to carefully read these instructions. The blue arrow will appear on the screen when you are able to move on, but you can always spend more time with the instructions if you would like."

Fig. A1. Existing Back Matter Text (ET) + Circle Visual Displays.



This makes it easier to find the lowest common denominator (LCD), the smallest number that the denominator can be, while keeping in mind that the denominator needs to be the same in all fractions when you add or subtract them.

$$\frac{1}{5} + \frac{6}{15} = ?$$



$$\frac{1}{5} + \frac{6}{15} \begin{matrix} \div 3 \\ \div 3 \end{matrix} = ?$$

$$\frac{1}{5} + \frac{2}{5} = \frac{3}{5} \quad (\text{The LCD is 5.})$$



As any fraction collector will tell you, reducing a fraction can be very handy. And who knows? It may be a very valuable fraction—in disguise!

Fig. A2. Researcher-Generated Back Matter Text (RGT) + Circle Visual Displays.

FRACTION EQUIVALENCE

Hey there! Let's talk all about how much fun we can have with equivalent fractions. When we say fractions are equivalent, we mean the fractions are the same. The two fractions might have different numbers on the top (numerator) and the bottom (denominator) of the line, but they can still be equal! Finding equivalent fractions will be helpful for when we add and subtract fractions later!

Let's see if these fractions are equivalent: $\frac{2}{10}$ and $\frac{1}{5}$

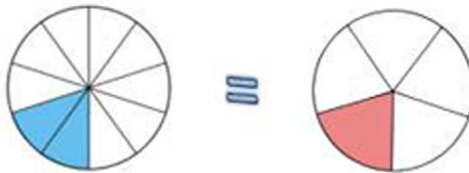
Let's say we have a circle that we cut into 10 equal sections, and we color 2 of those sections blue. That shows us $\frac{2}{10}$.



Let's say we have another circle that we cut into 5 equal sections, and we color 1 of those sections **red**. That shows us $\frac{1}{5}$.



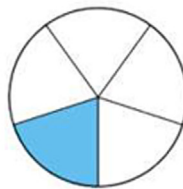
We can see that these two fractions take up the exact same space in each circle. That means they are equivalent! $\frac{2}{10}$ is the same as $\frac{1}{5}$.



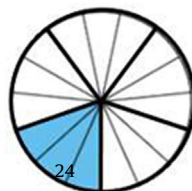
Now let's use those skills to add fractions together!

Let's try and add: $\frac{1}{5} + \frac{6}{15} = ?$

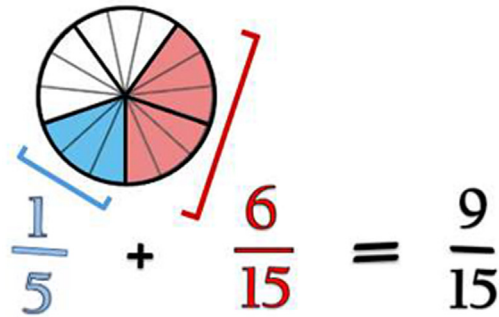
First, we will take a circle and cut it into 5 equal sections and color 1 of those 5 sections **blue**.



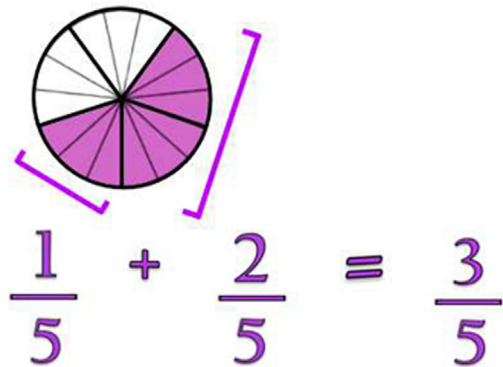
Next, we will take the same circle that is already partially colored blue and cut it into 15 equal sections (including the sections that were already colored).



Now, we will color 6 of the 15 uncolored sections **red**.



Finally, we will add the number of blue sections and red sections together to see how many **purple** sections of the total we have. All together, we have 9/15 of the sections colored in.



9/15 is equivalent to, or the same as, 3/5!

Fig. A3. Existing Back Matter Text (ET) + Number Line Visual Displays.

REDUCING FRACTIONS

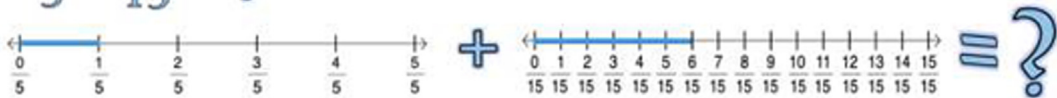
Simple—that’s the biggest little word in the fraction game. The simpler the fraction, the easier it is to add or subtract. In mathsppeak, the Reducer works by finding the greatest common factor (GCF), the largest number that both the numerator and the denominator can be divided by.

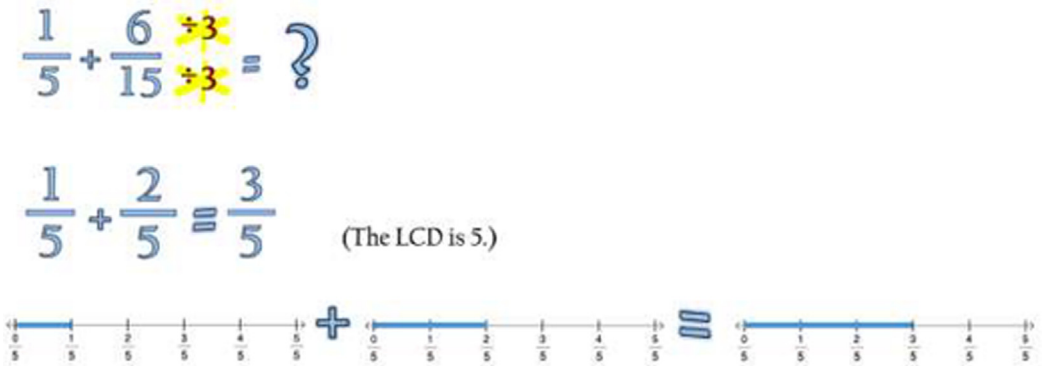
$\frac{2}{10} \div 2 = \frac{1}{5}$ (The GCF is 2.)



This makes it easier to find the lowest common denominator (LCD), the smallest number that the denominator can be, while keeping in mind that the denominator needs to be the same in all fractions when you add or subtract them.

$\frac{1}{5} + \frac{6}{15} = ?$





As any fraction collector will tell you, reducing a fraction can be very handy. And who knows? It may be a very valuable fraction—in disguise!

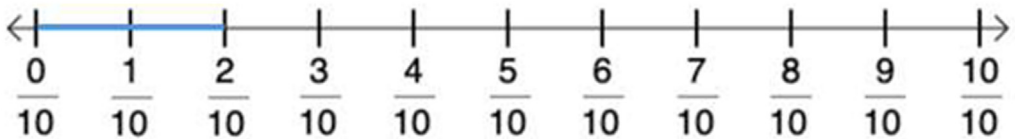
Fig. A4. Researcher-Generated Back Matter Text (RGT) + Number Line Visual Displays.

FRACTION EQUIVALENCE

Hey there! Let's talk all about how much fun we can have with equivalent fractions. When we say fractions are equivalent, we mean the fractions are the same. The two fractions might have different numbers on the top (numerator) and the bottom (denominator) of the line, but they can still be equal! Finding equivalent fractions will be helpful for when we add and subtract fractions later!

Let's see if these fractions are equivalent: $\frac{2}{10}$ and $\frac{1}{5}$

Let's say we have a number line that we cut into 10 equal sections, and we color 2 of those sections blue. That shows us $\frac{2}{10}$.



Let's say we have another number line that we cut into 5 equal sections, and we color 1 of those sections **red**. That shows us $\frac{1}{5}$.



We can see that these two fractions take up the exact same space on each number line. That means they are equivalent! $\frac{2}{10}$ is the same as $\frac{1}{5}$.



Now let's use those skills to add fractions together!

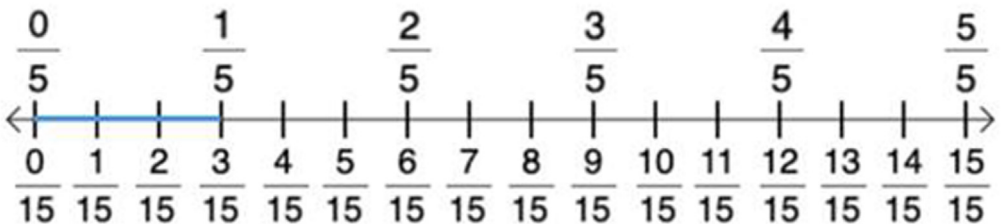
Let's try and add:

$$\frac{1}{5} + \frac{6}{15} = ?$$

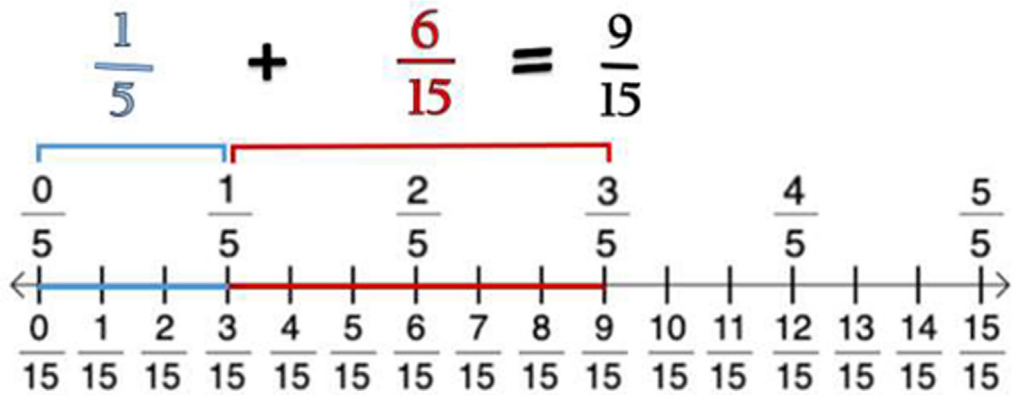
First, we will take a number line and cut it into 5 equal sections and color 1 of those 5 sections **blue**.



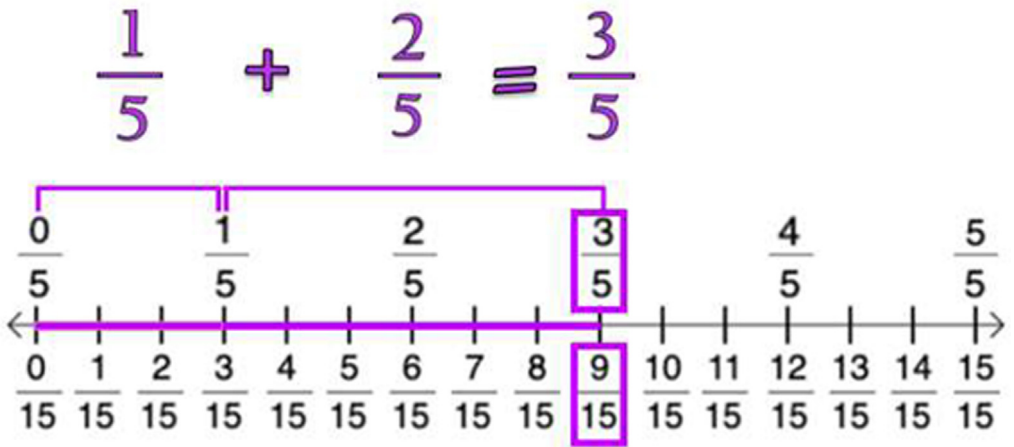
Next, we will take the same number line that is already partially colored blue and cut it into 15 equal sections (including the sections that were already colored).



Now, we will color 6 of the 15 uncolored sections **red**.



Finally, we will add the number of blue sections and red sections together to see how many **purple** sections of the total we have. All together, we have 9/15 of the sections colored in.



9/15 is equivalent to, or the same as, 3/5!

Note. For interpretation of the references to color in Appendix A, the reader is referred to the Web version of this article.

Appendix B. Fraction stimuli

Table B1

Fraction equivalence problems used in the study.

Fractions presented	Correct answer
$9/21 = 15/35$	True
$8/10 = 20/25$	True
$4/9 = 36/81$	True
$4/9 = 35/36$	False
$12/72 = 6/7$	False
$5/8 = 27/45$	False
$1/4 = 3/5$	False
$6/15 = 2/5$	True

Table B2

Fraction arithmetic questions used in the study.

Fraction arithmetic problem presented	Correct answer (or equivalent fraction)
$2/3 + 3/5 =$	19/15
$3/5 + 4/5 =$	7/5
$2/3 - 3/5 =$	1/15
$4/5 - 3/5 =$	1/5
$3/5 + 1/5 =$	4/5
$3/5 - 1/5 =$	2/5
$3/5 - 1/4 =$	7/20
$1/4 + 3/5 =$	17/20

Appendix C. Supplementary material

Supplementary material to this article can be found online at <https://doi.org/10.1016/j.jecp.2024.105983>.

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